

Bhadra Korukonda

Full Stack Developer

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Detail-oriented Computer Science student with hands-on experience building full-stack web applications and production-grade backend systems using Java and Spring Boot; developed scalable REST APIs, React frontends, and end-to-end systems using PostgreSQL, MongoDB, Docker, and CI/CD pipelines. Proven ability to ship clean, maintainable code and optimize performance in real-world projects.

SKILLS

Programming Languages: Python, Java, JavaScript
Libraries/Frameworks: PyTorch, scikit-learn, NumPy, Flask, React
Tools/Platforms: FastAPI, Flask, REST APIs, Git, Docker, Kubernetes, Prometheus, Grafana
Databases: PostgreSQL, MongoDB, SQL

PROFESSIONAL EXPERIENCE

Research Centre Imarat(RCI), DRDO
Data Scientist Intern | June 2024 – Aug 2024

- Automated Git workflows with Python, cutting version control overhead by **30%**.
- Developed statechart validation tools using OOP and tests, boosting accuracy by **25%**.
- Streamlined pipelines with SQL and Git CI, accelerating debugging and deployment.

IIT Hyderabad
Research Intern | Python, PyTorch Geometric, Pandas, Geopy, GRU/LSTM, GNNs (GCN, GAT, EGAT) May 2025 – Aug 2025

- Implemented a hybrid Edge-Aware Graph Attention Network (EGAT-GRU) to forecast PM2.5 levels across Delhi using spatio-temporal air quality data.
- Encoded wind direction and geodesic distance as edge features, enabling wind-informed pollutant transport learning and dynamic spatial attention.
- Achieved state-of-the-art performance (RMSE = 16.72, MAE = 10.05), outperforming GAT, GCN, and physics-based baselines for real-time pollution prediction.

PROJECTS

Expense Tracker | Full-Stack (Java + Spring Boot + React)

- Developed a complete personal finance tracker using Java 17, Spring Boot 3, Spring Data JPA (PostgreSQL) and MongoDB for flexible logs.
- Implemented RESTful CRUD APIs with validation, pagination, and custom SQL/NoSQL queries.
- Built responsive React frontend with form handling, real-time charts (Recharts), and Axios integration
- Containerized backend + frontend with Docker.

Cloud Automation & Monitoring Pipeline | Docker, Aws, Prometheus, Graphana, Jenkins, Python

- Engineered an automated CI/CD pipeline to streamline build, test, and deployment workflows using Docker and AWS.
- Integrated Grafana dashboards to visualize system metrics and deployment health in real time.
- Automated regression testing and environment provisioning, improving deployment reliability and reducing setup time by 40%.
- Implemented container orchestration principles to simulate distributed testing environments, mirroring real-world load scenarios.

Automated Resume Ranking System using NLP and MCDM | Python, Flask, spaCy, scikit-learn ,PyMCDM, Tesseract OCR

- Built end-to-end resume evaluation system with Flask REST APIs, real-time processing pipeline, and interactive React dashboard.
- Integrated Tesseract OCR + spaCy NLP pipeline to extract and structure resume data, enabling automated candidate scoring and ranking.
- Implemented TOPSIS/AHP algorithms to compute weighted candidate scores, improving shortlisting efficiency by ~40% and reducing human bias.
- Deployed modular, Docker-containerized application allowing HR users to upload resumes, adjust weights, and view ranked results instantly.

EDUCATION

Bachelor of Technology (CS Engg), Woxsen University, GPA: 3.0/4.0 Aug 2022 – July 2026
Narayana Jr College(SSC), Hyderabad, Percentage: 87.1% June 2020– May 2022
Chinmaya Vidyalaya(CBSE),Hyderabad Percentage: 90% May 2008 – June 2020